FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
CONCEPT RECAPITULATION TEST – IV
PAPER –2
TEST DATE: 26-04-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. D

Sol. From 2nd lens $\frac{1}{v}-\frac{1}{2\ell}=\frac{1}{-\ell}$ or $v=-2\ell$

$m_1=-1$

From 3rd lens $\frac{1}{v}-\frac{1}{-3\ell}=\frac{1}{2\ell}$ or $v=6\ell$

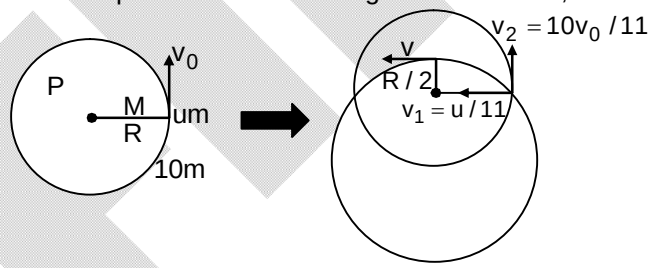
$m_2=-2$

$h_i=(m_1\times m_2)h_0$

$=2h$

2. A

Sol. A is the space station is moving in circular orbit,



$\frac{GM(10m)}{R^2}=\frac{(10m)v_0^2}{R}$

$\Rightarrow \quad v_0=\sqrt{\frac{GM}{R}}$ ....(i)

Let u be the velocity of meteorite.

Velocity of the space station after collision can be obtained from momentum conservation.

$mu=(10m+m)v_1 \quad \Rightarrow \quad v_1=\frac{u}{11}$ (perpendicular to meteorite's motion)

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$10m.v_0 = (10m+m)v_2 \Rightarrow v_2 = \frac{10}{11}v_0$ (parallel to meteorite's motion)

Let v be the velocity of space station at closest distance from angular momentum conservation

$10mv_0 \times R = 11\,mv\,\frac{R}{2} \Rightarrow v = \frac{20v_0}{11}$

From energy conservation

$$\frac{1}{2}\times(11\,m)\left(v_1^2+v_2^2\right)-\frac{GM(11\,m)}{R}=\frac{1}{2}\times(11\,m)v^2-\frac{GM.11m}{R/2}$$

$$\Rightarrow \left(\frac{u}{11}\right)^2+\left(\frac{10v_0}{11}\right)^2-\frac{2GM}{R}=\left(\frac{20v_0}{11}\right)^2-\frac{4GM}{R}$$

$$\Rightarrow \frac{u^2}{11^2}=\frac{400v_0^2}{11^2}-\frac{100v_0^2}{11^2}-\frac{2GM}{R}$$

$$\Rightarrow u^2=\frac{GM}{R}(400-100-242)=58\frac{GM}{R}$$

$$u=\sqrt{\frac{58GM}{R}}$$

3. B

Sol. Since the cell gives out a power of 10W, a current 2A must flow through the cell towards left.

$\therefore$ Power consumed in 2Ω resistor $= 2^2\times 2 = 8W$

Total current flowing in $1\Omega = 7$ Amp.

$\therefore$ Power consumed by $1\Omega = 7^2\times 1 = 49W$

4. D

Sol. All collision are elastic hence velocity will be exchanged.

For $0 < t < 4$ $\quad f_1 = f_0\left[\frac{340}{340-40}\right] = f_0\left[\frac{34}{30}\right]$

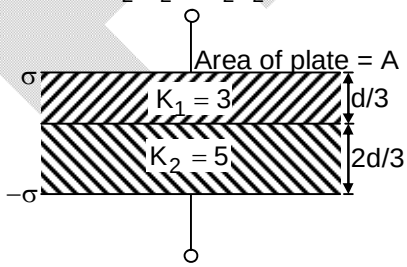
For $4 < t < 8$ $\quad f_2 = f_0\left[\frac{340-40}{340}\right] = f_0\left[\frac{30}{34}\right]$

For $8 < t < 12$ $\quad f_3 = f_0\left[\frac{340+40}{340}\right] = f_0\left[\frac{38}{34}\right]$

For $12 < t < 16$ $\quad f_4 = f_0\left[\frac{340}{340+40}\right] = f_0\left[\frac{34}{38}\right]$

5. AD

Sol. (i) $\frac{e_1}{e_2}=\frac{\in_1 E_1^2}{\in_2 E_2^2}=\frac{k_1E_1^2}{k_2E_2^2}=\left(\frac{k_1}{k_2}\right)\left(\frac{k_2}{k_1}\right)^2=\frac{k_2}{k_1}=\frac{5}{3}$



(ii) $\sigma_B=\sigma\left(1-\frac{1}{k_1}\right)-\sigma\left(1-\frac{1}{k_2}\right)=\sigma\left(\frac{1}{k_2}-\frac{1}{k_1}\right)=-\frac{2\sigma}{15}$

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6. ABC

Sol. Let N be the number of fringes within the length x, then we have,

$$\beta N = x \Rightarrow \frac{D\lambda}{d}N = x \Rightarrow N = \frac{xd}{\lambda D}$$

At any time t

$$N = \frac{x}{\lambda D}(d + vt)$$

$$\frac{dN}{dt} = \frac{xv}{\lambda D}.$$

7. AC

Sol. Deviation, $\delta = i + e - A$

dispersion, $\theta = (\delta_v - \delta_R) = (n_v - n_R)A$

$\delta_1 = -\delta_2$

$\delta_1 + \delta_2 = 0$

$\therefore$ Final ray will be parallel to initial ray.

8. ABD

Sol. Perimeter is decreasing at a rate of 2v

$$\therefore \quad \frac{d}{dt}(2\pi r) = 2v \Rightarrow \frac{dr}{dt} = \frac{v}{\pi}$$

$$\therefore \quad r = \left(r_0 - \frac{v}{\pi}t\right)$$

$$\phi = B \cdot \pi r^2 \Rightarrow \varepsilon = \left|\frac{-d\phi}{dt}\right| = B \cdot 2\pi \cdot r\frac{dr}{dt}$$

$$\therefore \quad \varepsilon = 2B\pi\left(r_0 - \frac{V}{\pi}t\right)\cdot\frac{v}{\pi} = 2Bv\left(r_0 - \frac{v}{\pi}t\right)$$

$$I = \frac{\phi}{R} = \frac{2Bvr}{\lambda \cdot 2\pi r} = \frac{Bv}{\pi\lambda}$$

9. ABC

Sol. $\tau = I\alpha$

$$mg\frac{l}{2} = \frac{ml^2}{3}\alpha$$

$$\Rightarrow \quad \alpha = \frac{3g}{2l}$$

Acceleration of CM of AC

$$= \alpha\frac{l}{4} = \frac{3g}{8}$$

Acceleration of CM of CB

$$= \alpha\frac{3l}{4} = \frac{9g}{8}$$

For part CB, $F + \frac{mg}{2} = ma_{cm}$

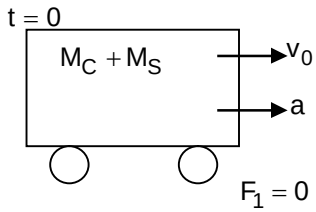
$$F = \frac{mg}{16}$$

$\therefore$ (A, B, C)

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10. AB

Sol.



$$F = \frac{mdv}{dt} + (v - v_0)\frac{dm}{dt}$$

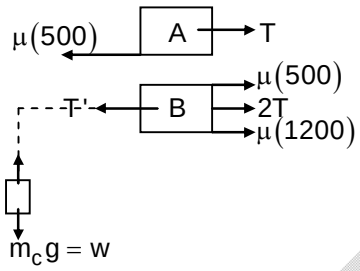
$$0 = \frac{mdv}{dt} + (v - v_0)\frac{dm}{dt}$$

$$\Rightarrow \frac{dv}{dt} = 0$$

Section – B

11. 9

Sol.



$3T + 0.3 \times 1200 = m_c g = W$ and $T = \mu(500) = 0.3 \times 500$

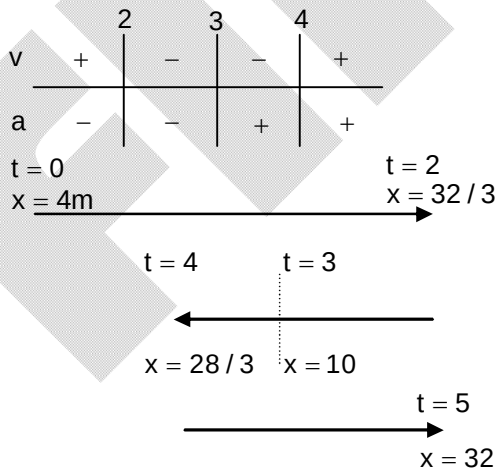
$W = m_0 g = 810$ N.

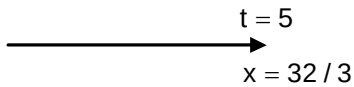
12. 4

Sol. $x = t^3/3 - 3t^2 + 8t + 4$

$v = t^2 - 6t + 8 = (t-2)(t-4)$

$a = 2(t-3)$





$$S_1 = \left(\frac{32}{3} - 4\right) + \left(\frac{32}{3} - \frac{28}{3}\right) + \left(\frac{32}{3} - \frac{28}{3}\right) = \frac{28}{3}\text{m}.$$

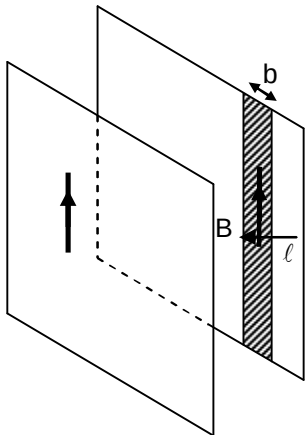
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$$S_2 = \left(\frac{32}{3} - 4\right) + \left(10 - \frac{28}{3}\right) = \frac{20}{3} + \frac{2}{3} = \frac{22}{3}\text{m}$$

$$\frac{S_1}{S_2} = \frac{28}{22} = \frac{14}{11} = \frac{3\alpha + 2}{11} \Rightarrow \alpha = 4$$

13. 2

Sol. Magnetic field due to one of the sheet



$B = \frac{\mu_0 K}{2}$ Parallel to second sheet

Force on section of width b

$$F = bK\ell\frac{\mu_0 K}{2}$$

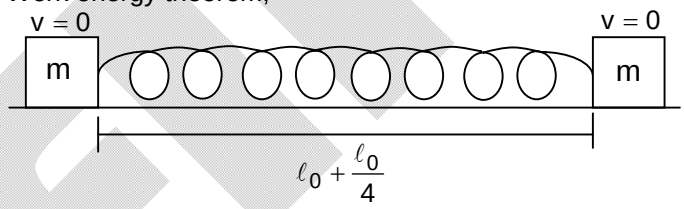
Force per unit area

$$P = \frac{B}{\ell b} = \frac{\mu_0 K^2}{2}$$

$$P = 4\pi \times 10^{-7}\frac{1}{2\pi}$$

14. 8

Sol. Work energy theorem,



$$-\mu mg\left(\ell_0 + \frac{\ell_0}{4}\right) = \frac{1}{2}K\left(\frac{\ell_0}{4}\right)^2 - \frac{1}{2}K\ell_0^2 \qquad \mu = \frac{3K\ell_0}{8mg}$$

15. 8

Sol. $N = mg\cos\theta + m\omega^2 R = mg\cos\theta + m\left(\frac{5g}{R}\right)R \Rightarrow N = 5\,mg + mg\cos\theta > 0$, so block does not leave circular motion

$f_r = ma_T$ For limiting case $\mu N = ma_T \quad \Rightarrow \quad \mu\left(mg\left(5 + \cos\theta\right)\right) = mg\sin\theta \Rightarrow \mu = \frac{\sin\theta}{5 + \cos\theta}$

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$$\frac{d\mu}{d\theta}=\frac{(5+\cos\theta)(\cos\theta)-\sin\theta(-\sin\theta)}{(5+\cos\theta)^2}=0\Rightarrow\cos\theta=-\frac{1}{5}$$

$$\mu_{max}=\frac{\sqrt{1-\frac{1}{25}}}{5-\frac{1}{5}}=\frac{2\sqrt{6}}{24}\Rightarrow x=8$$

16. 7

Sol. In an adiabatic expansion,

$TV^{\gamma-1}=$ constant

$$T_0V^{\gamma-1}=T\left(\frac{V}{5}\right)^{\gamma-1}\qquad\gamma=1+\frac{2}{5}$$

$$\therefore\quad T=(273).(5)^{2/5}$$

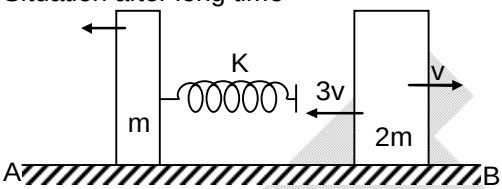
$$<(KE)_{rotational}>=kT$$

$$=1.38\times10^{-23}\times273\times(5)^{2/5}$$

$$=7\times10^{-21}J(approx).$$

17. 8

Sol. Situation after long time



$$\text{Work done}=\Delta K=\frac{1}{2}(2m)(v)^2-\frac{1}{2}(2m)(3v)^2=$$

$$=mv^2-9mv^2=-8mv^2.$$

18. 2

Sol. $\int_0^4 F_x dt=m(v_x-u_x)$

$$\&\int_0^4 F_y dt=m(v_y-u_y)$$

$$\therefore\vec{v}=-\hat{i}+11\hat{j}$$

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Chemistry

PART – II

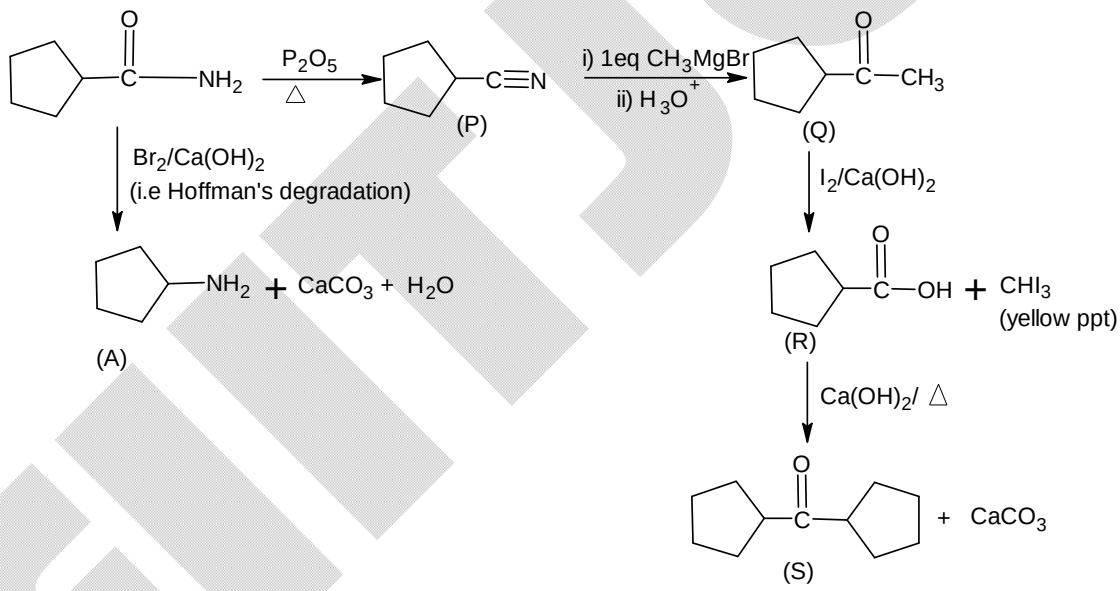
Section – A

19. A
Sol. $FeCl_3(aq)$ being acidic due to cationic hydrolysis, basic reagents can intensify the hydrolysis.

20. B
Sol. (X) = $[Co(SCN)_4]^{2-}$; (Y) = $[Ni(dmg)_2]$.

21. A
Sol. $X(g) + nY(g) \longrightarrow XY_n(g)$ $\Delta H_f^0 = \Delta H_f^0[XY_n(g)] - [\Delta H_f^0\{X\}(g) + n\Delta H_f^0\{Y\}(g)]$
$= -1100 - [275 + 80n]$
$= -1375 - 80\,n$
$\therefore$ average X – Y bond enthalpy = 309.2 kJ/mol
$\therefore$ $XY_n = -(309.2)\,n$
$(-309.2)n = -1375 - 80n$
$n = 6$
$\therefore$ $XY_n \Rightarrow XY_6$

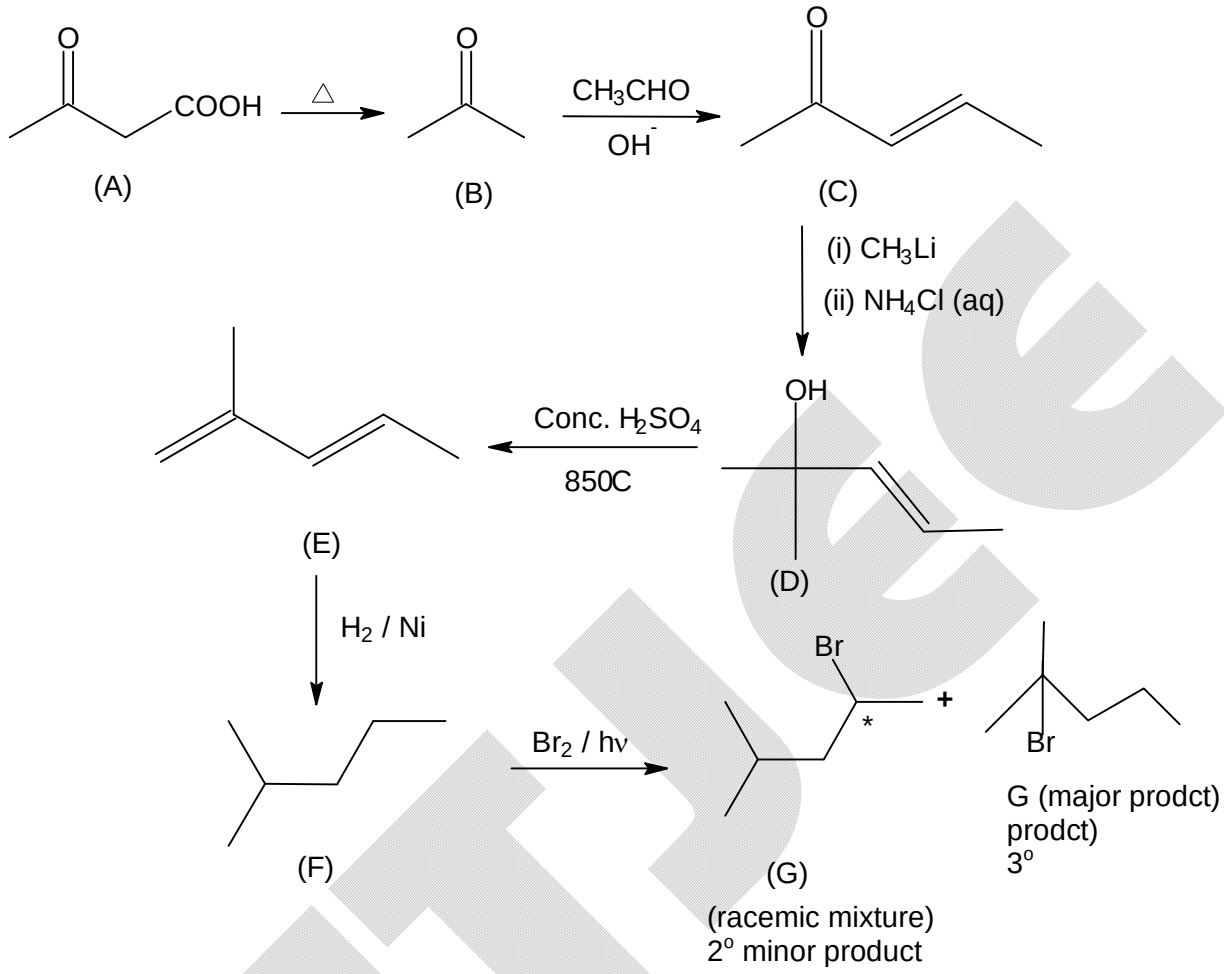
22. D
Sol.



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23. AB

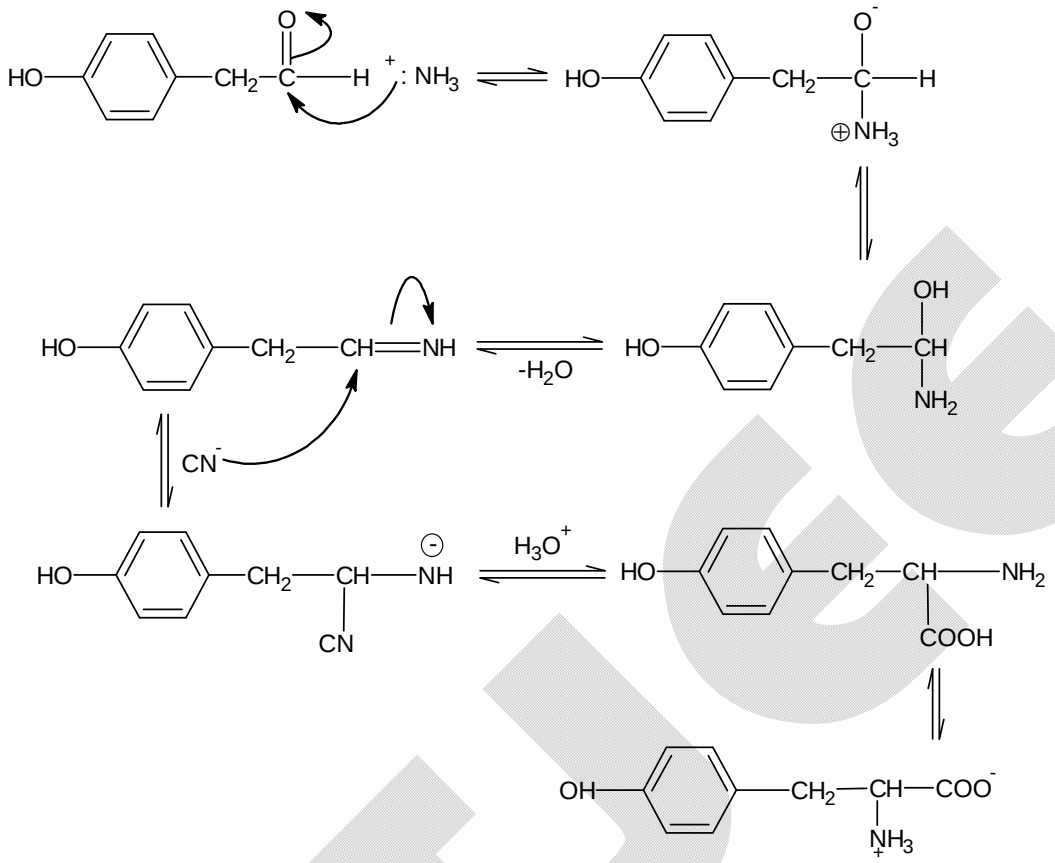
Sol.



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24. AB

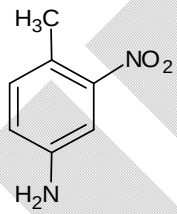
Sol.



25. A

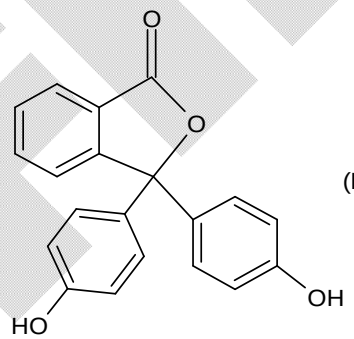
Sol. (A) Reduction of - NO_2 group.

(B)



will be the product.

(C)



(Phenolphthalein) will be the product.

26. ABC

Sol. (i) $\because \Delta V = 0$

$\therefore w = 0, q = \Delta U, \Delta H = C_p (T_2 - T_1), \Delta U = C_V (T_2 - T_1)$

(ii) $w = 0, \Delta U = 0, \Delta H = 0, q = 0$

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(iii) $-w = P\Delta V = P(V_2 - V_1)$
$= R(T_2 - T_1) = -831.4$
$\therefore w = 831.4\ Jmol^{-1}$
$\Delta U = C_V \Delta T$
$\Delta H = C_P \Delta T$
$q = C_V \Delta T - P\Delta V$
Total $\Delta U = 0$
Total $\Delta H = 0$
$w = +831.4$ joule mol^{-1}
$q = -831.4$ joule mol^{-1}

27. ABCD

Sol. (A) Carbon has pronounced ability to form $P\pi - P\pi$ multiple bonds to itself and to other elements like O and N. (Correct)
(B) In diamond each carbon atom is linked tetrahedrally to four other carbon atoms by sp^3 hybridisation. (Correct)
(C) Graphite has planar hexagonal layers of carbon atoms held together by weak Vander Waal's forces (Correct)
(D) Out of CO_2, SiO_2, SnO_2 and PbO_2. CO_2, SiO_2 are acidic, SnO_2 is amphoteric and PbO_2 is an oxidizing agent. (Correct)

28. ABC

Sol. Separation is equivalent to edge length of fcc unit cell which is equal to $2\sqrt{2}\,r$.

Section – B

29. 0

Sol. $E^o = E^o_{Ag^+/Ag} + 0.0591 \log K_d^o$

$E^o_{NH_3/Ag(NH_3)_2^+/Ag} = 0.8 + 0.0591 \log 10^{-8}$

$= 0.3272\ V$ ..... (I)

$E^o_{CN^-/Ag(CN)_2^-/Ag} = 0.8 + 0.0591 \log 10^{-19.072}$

$= -0.3272\ V$ ..... (II)

(I) + (II) = 0

30. 9

Sol. Factual

31. 4

Sol. $6CaOCl_2 \xrightarrow{\Delta} Ca(ClO_3)_2 + 5CaCl_2$

$2NaCl + H_2SO_4 \longrightarrow Na_2SO_4 + 2HCl$

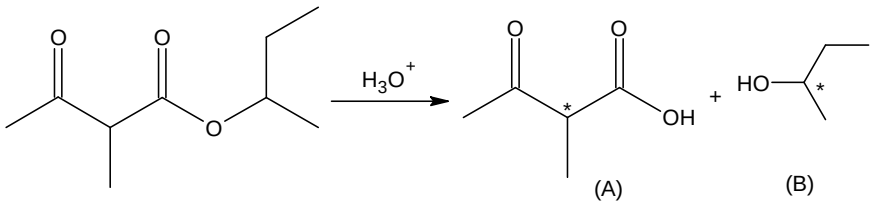
$Na_2S_2O_3 + 2HCl \longrightarrow 2NaCl + H_2O + S\downarrow + SO_2$

$\therefore a = 1, b = 5, c = 2, d = 2, e = 2$
$(a + b + c) - (d + e) = (1 + 5 + 2) - (2 + 2)$
$= 4$

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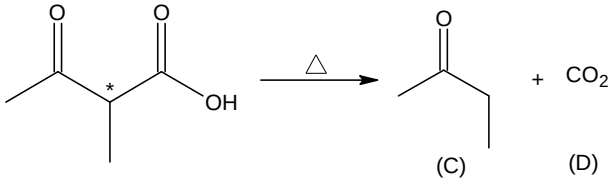
32. 1

Sol.



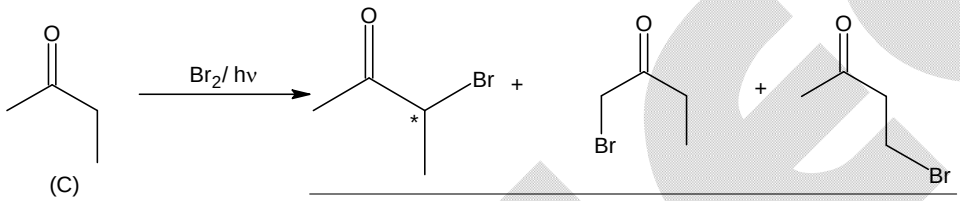
(A) (B)

Retention in configuration in both (A) and (B)



(C) (D)

In (C) and (D) no chiral carbon present.



(C) (E)

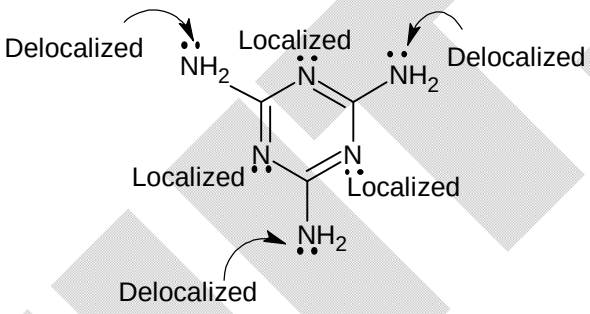
Chiral carbon only in the 1st compound formed from (C). This will exist in the enantiomeric pair.

∴ Sum of total number of optically active products = 4.

But total number of enantiomeric pairs = 1

33. 3

Sol.



34. 6

Sol. PbS, CuS, HgS, Ag_2S, NiS, CoS are black coloured sulphide

MnS is pink coloured

ZnS is white coloured

SnS_2 is yellow coloured

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35. 7

Sol.

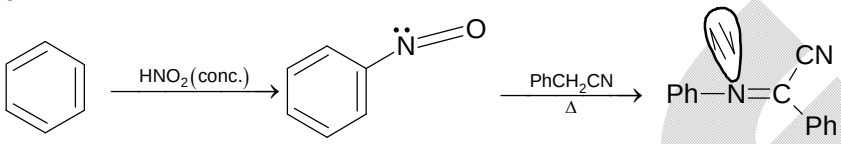
(4) ← $C_4H_9-N(CH_3)-CH_3$

(2) ← $C_3H_7-N(C_2H_5)-CH_3$

(1) ← $C_2H_5-N(C_2H_5)-C_2H_5$

36. 9

Sol.



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Mathematics

PART – III

Section – A

37. B

Sol. Let A = event that balls are of different colors

B = event that one ball is red and other is white

$$P\left(\frac{B}{A}\right)=\frac{P(A\cap B)}{P(A)}=\frac{\frac{{}^{6}C_1\times{}^{3}C_1}{{}^{13}C_2}}{\frac{{}^{3}C_1\times{}^{6}C_1+{}^{3}C_1\times{}^{4}C_1+{}^{4}C_1\times{}^{6}C_1}{{}^{13}C_2}}=\frac{6\times3}{6\times3+3\times4+4\times6}=\frac{1}{3}$$

38. B

Sol. Each element has 7 ways to be placed in the circle.

All elements can be placed in 7^n ways.

39. C

Sol. $a+b+c=0$ .... (1)

$50<49a+7b+c<60 \Rightarrow 50<48a+6b<60$

$$\Rightarrow \frac{50}{6}<8a+b+10$$

$\Rightarrow 8a+b=9$ .... (2)

$70<64a+8b+c<80$

$\Rightarrow 70<63a+7b<80$

$$\Rightarrow 10<9a+b<\frac{80}{7}$$

$\Rightarrow 9a+b=11$

From (1), (2) and (3), a = 2, b = -7, c = 5

$\Rightarrow f(x)=2x^2-7x+5$

$\therefore f(10)=135 \Rightarrow k=10$

40. C

Sol. Let $f(x)=(x-x_1)(x-x_2)(x-x_3)(x-x_4)$

$$\Rightarrow |f(i)|=\sqrt{1+x_1^2}\sqrt{1+x_2^2}\sqrt{1+x_3^2}\sqrt{1+x_4^2}=1$$

$\Rightarrow x_1=x_2=x_3=x_4=0$

$\Rightarrow$ all four roots are zero

$\Rightarrow f(x)=x^4 \Rightarrow a+b+c+d=0.$

41. AB

Sol. Let P and Q be the points (α,β) and (α_1,β_1)

$\Rightarrow$ equations of AB and CD are $\frac{x}{a}\alpha+\frac{y}{b}\beta=1$ and $\frac{x}{a}\alpha+\frac{y}{b}\beta=1$ and $\frac{x}{a}\alpha_1+\frac{y}{b}\beta_1=1$

Also $\frac{\alpha^2}{a^2}+\frac{\beta^2}{b^2}=\frac{\alpha_1^2}{a^2}+\frac{\beta_1^2}{b^2}$

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$\Rightarrow \frac{\alpha}{\alpha_1} = \frac{\beta}{\beta_1} = -1$

$\Rightarrow$ PQ passes through origin and is bisected at the origin

42. BD

Sol. m and m^2 are roots of equation $\left(\frac{y}{x}\right)^2 - 6\left(\frac{y}{x}\right) + a = 0$

$m^3 = a$

$\Rightarrow m^3 = m^6 + 3m^3\left(m + m^2\right) = 216$

$a + a^2 3a.\ 6 = 216$

$a^2 + 19a - 216 = 0 \Rightarrow a = 8, a = -27$

43. CD

Sol. $\left(\alpha, \alpha^2\right)$ lie on the parabola $y = x^2$

$\left(\alpha, \alpha^2\right)$ must lie between the asymptotes of

hyperbola $\frac{x^2}{1} - \frac{y^2}{4} = 1$ in 1st and 2nd quadrant.

$\therefore$ asymptotes are $y = \pm 2x$.

$\Rightarrow 2\alpha < \alpha^2$

$\Rightarrow \alpha < 0$ or $\alpha > 2$.

and $-2\alpha - a^2$

$\alpha < -2$ or $\alpha > 0$.

$\therefore \alpha \in (-\infty, -2)$ or $\alpha \in (2, \infty)$.

44. AD

Sol. We have $r\left(r_1r_2 + r_2r_3 + r_3r_1\right) = \frac{\Delta^3}{(s-a)(s-b)(s-c)}$ and $r_1r_2r_3 = \frac{\Delta^3}{(s-a)(s-b)(s-c)}$

$\therefore x^2 - r\left(r_1r_2 + r_2r_3 + r_3r_1\right)x + r_1r_2r_3 - 1 = 0$ is satisfied by x = 1.

So, one root is x = 1 and other root is therefore $r_1r_2r_3 - 1$.

45. CD

Sol. $b^2 = (\text{semi} - \text{minor axis})^2 = \frac{2+3-2}{\sqrt{2}} \cdot \frac{3+5-2}{\sqrt{2}}$

$= \frac{3 \times 6}{2} = 9$

b = 3

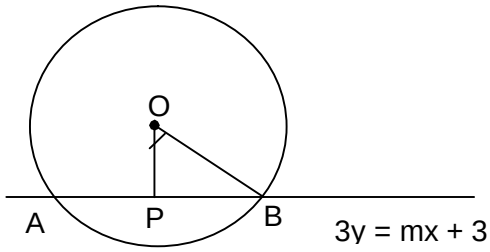
$b^2 = a^2 - a^2 - e^2, 2ae = \sqrt{5}$

$\Rightarrow e = \sqrt{\frac{5}{41}} \cdot a = \frac{\sqrt{41}}{2}$

46. BD

Sol. $OP = \frac{1}{\sqrt{2}}$

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$$\left|\frac{0-0+3}{\sqrt{9+m^2}}\right| = \frac{1}{\sqrt{2}}$$

$9+m^2 = 18$

$m^2 = 9 \Rightarrow m = \pm 3$

$m + 3 = 6, 0$.

Section – B

47. 6

Sol. Let the equations of tangents to the given circle and the ellipse respectively be

$$y = mx + 4\sqrt{1+m^2}$$

and $y = mx + \sqrt{25m^2+4}$

Since both of these represent the same common tangent,

$4\sqrt{1+m^2} = \sqrt{25m^2+4}$

$\Rightarrow 16\left(1+m^2\right) = 25m^2 + 4$

$\Rightarrow m = \pm\frac{2}{\sqrt{3}}$

The tangent is at a point in the first quadrant $\Rightarrow m < 0$.

$\Rightarrow m = -\frac{2}{\sqrt{3}}$, so that the equation of the common tangent is

$y = -\frac{2}{\sqrt{3}}x + 4\sqrt{\frac{7}{3}}$

It meets the coordinate axes at $A\left(2\sqrt{7}, 0\right)$ and $B\left(0, 4\sqrt{\frac{7}{3}}\right)$

$\Rightarrow AB = \frac{14}{\sqrt{3}}$

48. 3

Sol. $\because x^2 + 4y^2 = 4$ or $\frac{x^2}{2^2} + \frac{y^2}{1^2} = 1$ .....(1)

$x^2 + 2y^2 = 6$ or $\frac{x^2}{\left(\sqrt{6}\right)^2} + \frac{y^2}{\left(\sqrt{3}\right)^2} = 1$ .....(2)

Let $R(\alpha, \beta)$ be the point of intersection of tangent to ellipse (2) at P and Q, then PQ will be the chord of contact of R.

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$\therefore$ Equation is $\frac{x\alpha}{6}+\frac{y\beta}{3}=1$ or $y=\frac{-\alpha}{2\beta}x+\frac{3}{\beta}$

$\because$ It is tangent to $x^2+4y^2=4$

$\therefore \left(\frac{3}{\beta}\right)^2=4\cdot\frac{\alpha^2}{4\beta^2}+1 \qquad \left[\because c^2=a^2m^2+b^2\right]$

$\frac{9}{\beta^2}=\frac{\alpha^2+\beta^2}{\beta^2}$ or $\alpha^2+\beta^2=9$

$\Rightarrow (\alpha,\beta)$ lie on director circle $x^2+y^2=\left(\sqrt{6}\right)^2+\left(\sqrt{3}\right)^2$

i.e. $x^2+y^2=9$

$\therefore$ tangent at P and Q to ellipse $x^2+2y^2=6$ are at right angle.

49. 2

Sol. Let the coordinates of the points A and B be $A=\left(at_1^2, 2at_1\right), B=\left(at_2^2, 2at_2\right)$

The equation of the normal at A is

$y-2at_1=-t_1\left(x-at_1^2\right)$

As it passes through B, $2at_2-2at_1=-t_1\left(at_2^2-at_1^2\right)$

or $2a\left(t_2-t_1\right)=-at_1\left(t_2+t_1\right)\left(t_2-t_1\right)$

$\Rightarrow 2=-t_1\left(t_2+t_1\right)$ … (1)

Since OA and OB are perpendicular to each other

$\therefore -t_1t_2=4$

or $t_2=-\frac{4}{t_1}$

Substituting the value of t_2 in (1), we get $2=-t_1\left(\frac{-4}{t_1}+t_1\right)=4-t_1^2$ or $t_1^2=2$

$\therefore$ slope of AB = $\frac{2}{t_1+t_2}=\frac{-2t_1}{2}=\pm\sqrt{2}$

50. 9

Sol. Analyse integer and fractional part of $R=\left(15+\sqrt{220}\right)^{19}+\left(15+\sqrt{220}\right)^{82}$

51. 5

Sol. Use h(x) = f(x) – 10x

52. 2

Sol. $f(t)=144^t+144^{1-t}$, where $t=\sin^2 x, x\le t\le 1$

$f'(t)=\ln 144\left(144^t-144^{1-t}\right)$, we have $f'(t)=0$ at $t=\frac{1}{2}$, $f(t)$ increasing in $\left(\frac{1}{2},1\right)$ and decreasing in $\left(0,\frac{1}{2}\right)$ and $f(t)\Big|_{min}=f\left(\frac{1}{2}\right)=24$ and $f(t)\Big|_{max}$ at $f(0)=f(1)=145$

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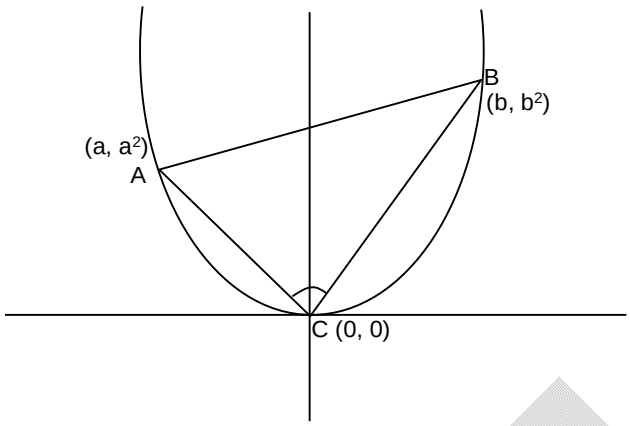
53. 6

Sol. $|A| = 2,\ |B| = 3,\ |C| = 5$

$$\det\left(A^2BC^{-1}\right) = \left|A^2BC^{-1}\right| = \frac{|A|^2|B|}{|C|} = \frac{4\times 3}{5} = \frac{12}{5}$$

54. 2

Sol. $AC \perp BC$



$$\frac{b^2}{b}.\frac{a^2}{a} = -1 \Rightarrow ab = -1 \qquad .....(1)$$

$a < 0. b > 0$

$$1 = \frac{1}{2}AC\cdot BC \Rightarrow AC\cdot BC = 2$$

$$\Rightarrow \sqrt{a^2+a^4}\sqrt{b^2+b^4} = 2 \,using\,(1)$$

$$\Rightarrow \sqrt{\frac{1}{b^2}+\frac{1}{b^4}}\sqrt{b^2+b^4} = 2 \Rightarrow \sqrt{2+\frac{1}{b^2}+b^2} = 2$$

$$\Rightarrow b^2+\frac{1}{b^2} = 2 \Rightarrow b = \pm 1$$

$\because b > 0, b = 1$

Hence B(1, 1)

answer 1 + 1 = 2

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